

Big Data Analytics
CISELEC 501A-CS41S1

Arias, Kenneth B.

Prof: John Richard Kho
08/16/2025

WEKA Understanding

OBJECTIVE: The objective of this exercise is to gauge your familiarity with using WEKA as a tool for evaluating data or performing data mining (AI) tasks. **SUBMISSION:** Group of three or individual.

QUESTIONS:

1. **When the WEKA Explorer is first started, which section tab is active, and what is its primary purpose?**
 - The section tab that was active when you first started weka explorer is the Preprocess tab. Its main purpose of this section is to help users familiarize with the loaded dataset. It highlights important details and information about the datasets that can help users to identify relevant information.
2. **List at least three different methods by which a user can load data into the Explorer's "Preprocess" section.**
 - The three different methods that can be used to load data are:
Loading a dataset by opening a File.
Loading a dataset through URL
Loading a dataset via database
3. **In the "Preprocess" panel, when an attribute is selected, what are three types of statistical information (e.g., counts or distributions) displayed in the "Selected attribute" box?**
 - Three types of statistical information that are displayed when selecting an attribute are:
No (List of non-unique values)
Label (The output value of a non-unique value of that selected attribute)
Count (The number of times a value has appeared throughout that attribute)
4. **Describe two distinct "Test options" available for evaluating a classifier in the Explorer's "Classify" section.**
 - Two distinct that are available for evaluating a classifier are:
 1. **Cross Validation** - This is where the models are divided into 'k' equal sized subsets or what we call "Folds". k represents the number of times should we divide the model. Per iterations, 1 of the folds is used as a testing case/set while the remaining 'k-1' folds are used to train.
 2. **Percentage Split** - This is where we divide the model into two. One is for testing and one is for training. The size of the division varies on the percentage of how much we want to split them.

5. **What is the main graphical display shown in the Explorer's "Visualize" section, and what key feature does it offer for exploring relationships between attributes?**

- The main graphical display that is shown in the Visualize section is a table, usually a scatter plot. It offers information about the error, prediction and original output value in relation between the attributes.

6. **Provide at least three available WEKA algorithms for the following tasks: Classification, Clustering, and Association.**

- Classification
 1. RandomForest
 2. RandomTree
 3. J48
- Clustering
 1. SimpleKMeans
 2. EM
 3. FarthestFirst
- Association
 1. Apriori
 2. FPGrowth
 3. FilteredAssociator

" I affirm that I have not given or received any unauthorized help in this assignment, and that this work is my own "

Kenneth Arias

